ArcLoom SPPU

Competitive Landscape Analysis

Ternary Computing, Clockless Perception, and Related Work

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Purpose

This document surveys all known ternary computing systems, clockless perception hardware, structural memory architectures, and related work. Its purpose is to establish that ArcLoom occupies a unique position in the landscape and to prepare clear distinctions for DARPA pitch, Efabless submission, and patent prosecution.

All ArcLoom claims reference validated demonstrations documented in the ASIC Architecture Specification v1.2.

Ternary Computing Hardware

Huawei Ternary Logic Gate (2025)

What: Patent CN119652311A for ternary logic gates using threshold voltage layering on CNTFET. Claims 40% fewer transistors, 60% less power.

Ternary type: Standard (0, 1, 2) — NOT balanced.

Hardware status: Patent filing. Unclear if silicon exists.

Computing model: Sequential instruction execution with 3-state gates.

ArcLoom distinction: Huawei improved the transistor. The computing model is unchanged — fetch/decode/execute with a clock. Standard ternary (0,1,2) lacks the symmetric negation property of balanced ternary (−1, 0, +1). The SPPU's combinational settling through signed coupling is architecturally impossible in Huawei's number system. (See ASIC Spec v1.2, §10.2.1.)

5500FP Ternary RISC Processor (2026)

What: 24-trit balanced ternary RISC processor on Efinix Trion T120F484 FPGA. 120-instruction ISA. Open hardware.

Ternary type: Balanced (−1, 0, +1).

Hardware status: Running on FPGA. No custom silicon.

Computing model: Von Neumann RISC — fetch, decode, execute, store.

ArcLoom distinction: The 5500FP treats trits as numbers in registers. ArcLoom treats them as coupled physical states in a field. The null trit in 5500FP is arithmetic zero. In SPPU, it is structural uncertainty — the system's admission that coupling pressure is insufficient to commit this dimension. The 5500FP validates that balanced ternary works on FPGA; ArcLoom builds a different computational model on top of it. (See ASIC Spec v1.2, §10.2.2.)

Microsoft BitNet b1.58 (2024–2026)

What: LLM with ternary weight quantization $\{-1, 0, +1\}$. 2B parameter model. Open-source inference engine.

Ternary type: Ternary VALUES on binary HARDWARE. Not ternary computing.

Hardware status: Runs on standard x86 CPUs. No custom hardware.

ArcLoom distinction: BitNet uses ternary as a compression trick — reducing model weight storage from 16-bit floats to 1.58-bit trits. The processor is entirely binary. The computation is still matrix multiplication (simplified to add/subtract). ArcLoom uses balanced ternary as the native computational substrate. The coupling fabric IS the computation. No matrix multiplication. No training. No inference.

TCN-CUTIE / CUTIE (ETH Zurich, 2022)

What: Fabricated ASIC (GlobalFoundries 22nm) for ternary neural network acceleration. 1,036 TOp/s/W efficiency.

Ternary type: Ternary weights, binary logic. The chip is a binary accelerator optimized for ternary-quantized weight matrices.

Hardware status: Real silicon. Production-grade.

ArcLoom distinction: TCN-CUTIE is a binary chip that processes ternary data. ArcLoom is a ternary chip that perceives through coupling. TCN-CUTIE requires trained neural network weights. ArcLoom derives weights from sensor physics with zero training.

TerEffic (2025)

What: FPGA architecture for ternary LLM inference. 16,300 tokens/second.

ArcLoom distinction: Ternary data on binary hardware with a trained model. ArcLoom is native ternary logic, no training.

Tiny Tapeout Balanced Ternary Calculator (USN Group)

What: 2-trit balanced ternary calculator (add/subtract/multiply) taped out on Skywater 130nm through Tiny Tapeout 03. Follow-up REBEL-2 ALU on TT05.

Ternary type: Balanced (−1, 0, +1). Binary-Encoded Ternary on CMOS.

Hardware status: Real silicon (Tiny Tapeout shuttle).

ArcLoom distinction: REBEL-2 does add/sub/mul. MathLoom does all of those PLUS folding division (41,430 tests, zero errors) with native free rounding via truncation. Nobody else has verified balanced ternary division on hardware. More fundamentally, REBEL-2 is a standalone ALU. MathLoom exists only at I/O boundaries of the SPPU — the perception path has no ALU at all.

Carbon Nanotube Ternary Logic (Science Advances, Jan 2025)

What: CNT source-gating transistors with native 3-state switching. Ternary inverters, NMIN/NMAX gates, ternary SRAM, ternary neural network.

Hardware status: Lab-scale fabricated devices. Not a system.

ArcLoom distinction: Individual ternary gates, not an architecture. Their ternary neural network is trained. ArcLoom is a complete perception system from sensor to motor, training-free.

USN Ternary Research Group (Univ. of South-Eastern Norway)

What: Academic research group building EDA tools for ternary VLSI. Mixed Radix Circuit Synthesizer (MRCS). Founded 2019.

Hardware status: EDA tools and simulation. No fabricated chips.

ArcLoom distinction: USN builds the tools to design ternary chips. ArcLoom IS a ternary chip (on FPGA, ASIC spec'd). Complementary, not competing. Potential collaboration for ArcLoom-256 tape-out.

Historical: Setun (Moscow State University, 1958)

What: First and only mass-produced balanced ternary computer. ∼50 units built. Discontinued for political reasons, not technical ones.

ArcLoom distinction: Setun was a general-purpose computer that happened to use balanced ternary. It validated the number system. It did not explore coupling, perception, or the null trit as structural uncertainty.

TCAM (Cisco, Broadcom — Production)

What: Ternary Content-Addressable Memory in network routers. Three states: 0, 1, X (don't care). Massively deployed.

ArcLoom distinction: TCAM's third state is a wildcard mask for pattern matching, not a computational state. TCAM does lookup, not logic. The "ternary" in TCAM has nothing in common with balanced ternary computing.

Three Generations of Ternary

Generation	Architecture	Innovation	Limitation
Huawei (2025)	Standard ternary gates	Better transistor	Same computing model
5500FP (2026)	Balanced ternary RISC	Better number system	Same computing model
ArcLoom (2026)	BT coupling fabric	Different computing model	—

Huawei improved the *transistor*. The 5500FP improved the *number system*. ArcLoom changed what *computation means*.

Clockless / Asynchronous Perception Hardware

Intel Loihi 2

What: 128 asynchronous neuron cores on Intel 4 process. Spiking neural network processor. Sensor fusion demonstrated (radar, vision, lidar).

ArcLoom distinction: Loihi runs trained spiking neural networks on async hardware. The hardware is clockless; the computation is ML. SPPU is combinational — no spikes, no neurons, no training, no iteration. The answer settles through wire propagation in ∼5 ns.

BrainChip Akida (AKD1500)

What: Event-driven digital processor. 800 GOPS at <300mW. Gesture recognition demonstrated with Prophesee event camera.

ArcLoom distinction: Akida runs Temporal Event-based Neural Networks (TENNs). Event-driven hardware, ML algorithms. SPPU has no neural network of any kind.

SpiNNaker2 (TU Dresden)

What: 153 ARM cores with async event routing. 22nm FDSOI. Deployed at Sandia National Labs.

ArcLoom distinction: SpiNNaker2 is a massively parallel ARM cluster for spiking neural network simulation. It has 153 clocked ARM cores connected by async routing. SPPU has zero clocked cores and zero neural networks.

Prophesee GenX320 (Event Camera)

What: 320×320 event camera where each pixel fires independently and asynchronously. 10,000 FPS equivalent. >140dB dynamic range.

ArcLoom distinction: Prophesee is an async *sensor*, not an async *processor*. Interesting as a future BSIL input source — event-driven pixels feeding directly into the SPPU's combinational fabric. Complementary technology.

Gap Analysis

Every existing clockless/async processor runs trained neural networks. The async fabric is the delivery mechanism; the computation is still ML.

Nobody has: purely combinational perception where the answer settles through coupling constraints with no spikes, no neurons, no weights learned from data, and no iteration. The SPPU is unique in this category.

Content-Addressable Structural Memory (Krimelack)

Existing CAM Technology

All deployed CAMs (TCAM in routers, memristor CAMs, ferroelectric CAMs) do bit-level or weight-level matching. They store addresses, keys, or feature vectors. None store **structural motifs** — coupled multi-field patterns that represent a settled loom state.

Memristor Habituation (Nature Communications, 2025)

What: Third-order memristor (HfO_2+TiO_x) with physical habituation. Robot arm ignores 71% of familiar stimuli while responding to threats. Non-volatile habituation survives power loss.

ArcLoom distinction: This is the closest existing work to Krimelack's habituation mechanism. The difference: memristor habituation is *analog*, *per-synapse*, and embedded in a trained SNN. Krimelack habituation is *digital*, *pattern-level* (full loom state), and operates through *combinational dead-zone feedback*. The memristor habituates individual connections. Krimelack habituates entire structural experiences.

Gap Analysis

No existing hardware combines:

1. Content-addressable structural motif storage
2. Cue-based recall from partial signatures
3. Pattern-level habituation via dead-zone feedback
4. Autonomous commit gating based on dimensional exhaustion (L6 structural lock)

Each piece exists in isolation somewhere. The combination is unique to Krimelack.

Balanced Ternary Arithmetic (MathLoom)

REBEL-2 (USN Group, Tiny Tapeout 05)

What: Balanced ternary ALU with MIN, MAX, ADD, SUB, MUL, CMP, SHFT. Submitted for ASIC fabrication on Skywater 130nm.

MathLoom distinction: REBEL-2 does not implement division. MathLoom's folding division — verified with 41,430 tests (integer, complex, matrix, polynomial) at zero errors — has no equivalent in any balanced ternary hardware. The native free rounding property (truncation = rounding in balanced ternary) is exploited as a design feature in MathLoom but is not used in REBEL-2's operations.

Georgia Tech Discrete ALU (Louis Duret-Robert)

What: 2-trit balanced ternary ALU from discrete CD4007 CMOS chips.

MathLoom distinction: Proof of concept on discrete components. Not integrated. No division. No verification suite.

MathLoom Unique Claims

1. Folding division with native free rounding — **No prior art**
2. Exact 1/3 representation exploited as design feature — **No prior art**
3. Computational completeness proven (complex, matrix, polynomial, trig)
4. 41,430 verification tests + 18 silicon tests at zero errors
5. Positioned as I/O boundary arithmetic, not central ALU

Structural Perception Without ML

Hyperdimensional Computing (UC San Diego, UC Berkeley, ETH Zurich)

What: Encodes sensor data into high-dimensional binary vectors. Does similarity matching via dot products. HyperSense (2024) does real-time raw sensor processing on FPGA. GraspHD (2024) does robotic grasping.

ArcLoom distinction: HDC encodes deterministically but classifies with a trained model. The encoding is training-free; the decision is not. SPPU is training-free end-to-end — from sensor to motor. HDC operates in high-dimensional binary space. SPPU operates in low-dimensional balanced ternary space where the null state is a first-class computational primitive.

Ternary Spike-Based SNN (Neural Networks, 2025)

What: Threshold-adaptive encoding converts analog to ternary spikes. QT-SNN processes with quantized ternary weights. Speech and EEG recognition.

ArcLoom distinction: Trained spiking neural network with ternary encoding. Not structural perception. Not combinational. Not training-free.

Gap Analysis

The following concepts return **zero results** in the literature:

- "Coupling weights" as a perception mechanism
- "Dimensional exhaustion" as a detection principle
- "Geometric inevitability" in hardware
- Perception that settles via coupling constraints without iteration or training
- 3^i positional weights as structural decoders in a perception fabric

ArcLoom's computational model — combinational settling through signed ternary coupling with dead-zone thresholding — has **No prior art**.

Dead Zone / Null State Computing

Three-Valued Logic

Kleene and Łukasiewicz three-valued logics are well-established in formal logic. Hardware implementations exist (TDDFET gates, memristive circuits, Huawei patent). In all cases, the third state represents "unknown" or "high-impedance" — a failure to resolve.

ArcLoom Distinction

In the SPPU, the null trit is not "unknown." It is "the coupling field has insufficient energy to collapse this dimension." This is an *active* computational state with specific consequences:

- It contributes zero to downstream coupling (multiplicative silence)
- It can be created by habituation (Krimelack dead-zone feedback)
- It can be created by insufficient input (sensor ambiguity)
- It is counted by L6 to detect dimensional exhaustion
- It can flip to committed ($+1$ or -1) when coupling pressure increases

No prior ternary architecture treats the null state as structural uncertainty with these computational properties.

Summary: What ArcLoom Has That Nobody Else Has

Capability	**Nearest Prior Art**	**Gap**
Combinational ternary perception	Loihi 2 (async SNN)	No ML, no spikes, no clock
3^i positional coupling weights	None	**No prior art**

Capability	Nearest Prior Art	Gap
Null trit as structural uncertainty	Three-valued logic	Not used as active computation
Krimelack structural motif memory	Memristor habituation	Not CAM, not pattern-level, not combinational
Folding division with native rounding	REBEL-2 (add/sub/mul only)	No division anywhere
L6 dimensional exhaustion	None	**No prior art**
Clockless habituation via dead-zone feedback	None	**No prior art**
Sensor-to-motor ternary chain	None	**No prior art**
Training-free embodied cognition	HDC (trains classifier)	End-to-end training-free
Autonomous obstacle navigation on ternary HW	None	**No prior art**

Potential Allies / Collaborators

Entity	What They Have	How They Help ArcLoom
USN Ternary Research Group	EDA tools (MRCS)	Synthesis and verification for tape-out
Prophesee	Event cameras (GenX320)	Async sensor input for SPPU
Efabless	Shuttle runs (SKY130, GF180)	ArcLoom-256 fabrication
Caltech/Yale async groups	ACT async design tools	Async BSIL timing verification

Anticipated Objections and Responses

"Isn't this just another ternary computer?"

No. The 5500FP is a ternary computer — fetch, decode, execute. ArcLoom has no instruction set. The answer settles through wire propagation. Different computational model entirely.

"How is this different from BitNet?"

BitNet uses ternary values on binary hardware. ArcLoom uses ternary logic as the native substrate. BitNet is a compression trick for trained LLMs. ArcLoom is a perception architecture that requires zero training.

"Huawei already made a ternary chip."

Huawei patented ternary gates using standard ternary (0,1,2). ArcLoom uses balanced ternary (-1,0,+1) which has the symmetric negation property required for signed coupling. Huawei's chip (if it exists) runs sequential instructions. ArcLoom settles combinationally. Different number system, different architecture.

"Neural accelerators already do low-power perception."

Neural accelerators require trained models, clocks, and sequential matrix multiplication. SPPU requires no training, no clock, and no multiplication in the decision path. Power is $10^9\times$ lower than ARM Cortex-M0 for equivalent perception.

"How do you know it works?"

19,683 arithmetic operations verified on FPGA silicon. 41,430 division tests at zero errors. Live sensor-to-motor chain demonstrated. 351-sample autonomous obstacle navigation with camera. Camera-sensor ambiguity correctly perceived. See ASIC Spec v1.2, §12 (Validation Status).

"What about Loihi / Akida / SpiNNaker?"

All three run trained spiking neural networks on async hardware. The hardware is clockless; the computation is ML. SPPU is purely combinational — no spikes, no neurons, no weights learned from data.

DSF-AI as ML Model Auditor (New Capability, May 2026)

During this competitive analysis, a new DSF-AI capability was discovered: the UF-Core kernel (L0–L4) can be run directly on the trained weight tensors of ML models. The kernel reads trained weights as structural signals and extracts coupling profiles per tensor, per layer, and per element.

Demonstration: CathodeX MACE Ensemble Audit

The DSF-AI kernel was run on the trained weights of the CathodeX MACE-MP-0 ensemble (5 members, 5.5M parameters each) — a materials science ML model for cathode screening. Findings:

1. **Ensemble members are structurally identical.** All 5 members produce identical DSF profiles to 6 decimal places. The ensemble's epistemic uncertainty estimate (inter-model variance) may measure noise, not genuine disagreement.
2. **Lithium has the weakest learned representation** of all cathode-relevant elements (coupling = 0.889, uncertainty = 0.434). The element most critical to cathode screening is the one the model knows least about.
3. **Parameter budget allocated to wrong elements.** Top 5 by weight magnitude: S, As, Ta, Sb, Nb — not cathode metals.

Implication: DSF-AI can audit any ML model by reading its weight structure. No other tool provides this capability. This is a natural extension of the kernel's domain-agnostic structural field extraction — trained weights are just another signal.

See *DSF-AI MACE Audit Report* (May 2026) for full details.
